Implementation of a Strength-Two Orthogonal Array Generator
for Software Testing

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Abstract

This report describes the implementation of an orthogonal-array generator for pairwise software testing. The work does not propose a new orthogonal-array construction; rather, it implements classical finite-field and product constructions in a form that accepts a number of factors and a common number of levels as input. Prime-power level counts are handled using a standard finite-field/projective-geometry construction, while composite level counts are handled by combining component arrays through a direct product. An independent verification procedure is included to check the defining pairwise-balance property of the generated array.

1 Introduction

Orthogonal arrays are classical combinatorial designs used in experimental design and combinatorial testing. Their theory, including constructions based on Galois fields, is well established [1, 2]. Finite projective geometry also gives a standard representation of strength-two orthogonal arrays [3].

The objective of this work is therefore *implementation*, not the introduction of a new construction. Given $k \geq 2$ factors and a common level count $s \geq 2$, the program constructs a strength-two orthogonal array, maps its symbols to software-test levels, and verifies the result computationally.

2 Orthogonal Arrays

An orthogonal array $\mathrm{OA}(N, k, s, 2)$ is an $N \times k$ array over s symbols such that, for every pair of columns, every ordered pair of symbols occurs the same number of times. If this common frequency is λ, then

$$N = \lambda s^2.$$

In software-testing terminology, the k columns are factors, the s symbols are their levels, and the N rows are test cases. Strength two corresponds to uniform pairwise coverage.

3 Construction Used

3.1 Prime-Power Number of Levels

Let the number of levels be a prime power $q = p^r$. The implementation uses the classical finite-field construction over $\mathrm{GF}(q)$ [2, 3].

Choose the smallest positive integer d satisfying

$$\frac{q^d - 1}{q - 1} \geq k. \tag{1}$$

There are $(q^d - 1)/(q - 1)$ one-dimensional subspaces of $\mathrm{GF}(q)^d$, so this choice provides at least k distinct projective directions.

Select representatives

$$v_1, \ldots, v_k \in \mathrm{GF}(q)^d$$

so that no two are scalar multiples. For every $x \in \mathrm{GF}(q)^d$, create one row and define

$$A[x, j] = x \cdot v_j. \tag{2}$$

The resulting array has $N = q^d$ rows.

Algorithm 1 Prime-power OA construction

Require: Number of factors k; prime power q

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Find the smallest d such that (q^d − 1)/(q − 1) ≥ k
Construct GF(q)
Choose k non-proportional vectors v_1, ..., v_k ∈ GF(q)^d
for all x ∈ GF(q)^d do
    for j ← 1 to k do
        A[x, j] ← x · v_j
    end for
end for
return A
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3.2 Composite Number of Levels

For an arbitrary s, write its prime-power factorization as

$$s = q_1 q_2 \cdots q_m,$$

where the q_i are pairwise coprime prime powers. A component OA is generated for each q_i. The components are then combined using the standard product construction for orthogonal arrays [2].

If A has s_A levels and B has s_B levels, corresponding symbols are encoded as

$$(a, b) \longmapsto a s_B + b.$$

Taking every pair of rows from A and B produces an array with $s_A s_B$ levels.

4 Correctness

Theorem 1. *For a prime power q, the construction in Equation (2) produces a strength-two orthogonal array.*

Proof. Consider two distinct columns corresponding to v_i and v_j. Since the chosen vectors are not scalar multiples, they are linearly independent. For any $(a, b) \in \mathrm{GF}(q)^2$, the equations

$$x \cdot v_i = a, \qquad x \cdot v_j = b$$

form two independent linear constraints on $x \in \mathrm{GF}(q)^d$. Hence they have exactly q^{d-2} solutions. Therefore every ordered pair (a, b) occurs exactly q^{d-2} times in every pair of columns, which is precisely the strength-two orthogonality condition. □

Algorithm 2 General OA generation

Require: Integers $k \geq 2$ and $s \geq 2$

Factor $s = q_1 q_2 \cdots q_m$ into prime powers
for $i \leftarrow 1$ to m **do**
  $A_i \leftarrow \textsc{PrimePowerOA}(k, q_i)$
end for
$A \leftarrow A_1$
for $i \leftarrow 2$ to m **do**
  $A \leftarrow \textsc{DirectProduct}(A, A_i)$
end for
Relabel symbols as $1, \ldots, s$
return A

The direct product preserves this property: if a pair occurs λ_A times in one component and λ_B times in the other, the corresponding combined pair occurs $\lambda_A \lambda_B$ times. Repeated products therefore preserve strength two.

5 Implementation and Verification

The implementation requires four principal components:

1. integer factorization of s into prime powers;
2. arithmetic in $\mathrm{GF}(q)$, using modular arithmetic for prime q and polynomial arithmetic modulo an irreducible polynomial for $q = p^r$;
3. generation of canonical representatives of one-dimensional subspaces; and
4. direct-product composition of component arrays.

The generated array is verified independently. For every pair of columns, the implementation counts all s^2 ordered pairs. With N rows, every count must equal

$$\lambda = \frac{N}{s^2}.$$

Thus verification directly checks the definition of a strength-two OA and helps detect implementation errors in finite-field arithmetic or array construction.

6 Scope and Limitations

The program assumes that every factor has the same number of levels and generates only strength-two arrays. It also does not claim to minimize the number of rows. The finite-field and product methods are used because they provide a systematic constructive route to a valid OA; a smaller orthogonal array, or a smaller pairwise *covering array*, may exist for the same testing problem.

Consequently, the contribution of this work is a software implementation and integration of established constructions, together with automatic parameter handling and verification, rather than a new result in combinatorial design theory.

7 Conclusion

A general strength-two orthogonal-array generator was implemented by combining established finite-field and direct-product constructions. The method accepts a number of factors and a common number of levels, determines suitable construction parameters, generates the test array, and verifies pairwise balance. This provides a compact mathematical basis for an Orthogonal Array Testing tool while clearly separating the implementation contribution from the classical theory on which it is based.

References

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